CHEZE GRINDS

Leaving Cert Examination

Mathematics

Paper 1

Higher Level

300 marks

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Instructions

There are **two** sections in this examination paper.

Section A	Concepts and Skills	150 marks	6 questions
Section B	Contexts and Applications	150 marks	4 questions

Answer **any five** questions from Section A.
Answer **any three** questions from Section B.

Write your Examination Number in the box on the front cover.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write all answers into this booklet. There is space for extra work at the back of the booklet. If you need to use it, label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

In general, diagrams are not to scale.

You will lose marks if your solutions do not include relevant supporting work.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here: _____

Section A	Concepts and Skills	150 marks

Answer **any five** questions from this section.

Question 1 **(30 marks)**

(a) Solve the following inequality for $x \in \mathbb{R}$:

$$|4x - 3| > 9$$

(b) $(3x + 2)$ is a factor of $6x^3 + x^2 - 19x - 6$.
Find the three real solutions of:

$$6x^3 + x^2 - 19x - 6 = 0$$

(c) Write $g(x) = 2x^2 - 12x + 7$ in the form $a(x - h)^2 + k$, where $a, h, k \in \mathbb{Q}$.
Write down the minimum value of $g(x)$ and the value of x at which it occurs.

Question 2 **(30 marks)**

(a) A function f is defined for $x \in \mathbb{R}$ by:

$$f(x) = 4x^3 - 3\sin 2x + e^x$$

(i) Find $f'(x)$.

(ii) The line ℓ is tangent to the curve $y = f(x)$ at the point where $x = 0$.
Find the equation of ℓ, giving your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.

(b) A function g is defined for $x \in \mathbb{R}$ by:

$$g(x) = \frac{3x^2 - 5}{7}$$

Find $g'(x)$. You do not need to simplify your answer.

(c) $k \in \mathbb{R}$ is a constant, and:

$$\int_1^k (4x - 2)\, dx = 12$$

Find the value of k.

Question 3 **(30 marks)**

(a) Find each of the following:

(i) $\int \left(6x^2 - 5 + 4\sin x\right) dx$

(ii) $\int \left(3e^{2x} + \frac{2}{x}\right) dx$

(b) The curve $y = f(x)$ passes through the point $(2, 10)$.
Given that $f'(x) = 3x^2 - 4x + 1$, find $f(x)$.

(c) Find the average value of the function $h(x) = x^2 - 3x + 5$ on the interval $[1, 4]$.
The average value of f on $[a, b]$ is $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$.

Question 4 **(30 marks)**

In this question, $i^2 = -1$.

(a) Write the following in the form $a + bi$, where $a, b \in \mathbb{R}$:

$$\frac{2 + 5i}{3 - 4i}$$

(b) Use de Moivre's theorem and the expression $(\cos\theta + i\sin\theta)^3$ to prove that:

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

(c) Use de Moivre's theorem to find two values of z for which:

$$z^4 = -8 + 8\sqrt{3}\,i$$

Give each answer in the form $a + bi$, where $a, b \in \mathbb{R}$. Leave surds in your answer.

Question 5 **(30 marks)**

(a) p is a positive constant. Use the laws of logarithms to write the following expression in the form $c + d \ln p$, where $c, d \in \mathbb{Z}$:

$$\ln\left(\frac{e^5 \cdot p^3}{e^2 \cdot p}\right)$$

(b) Solve for x, where $x \in \mathbb{R}$:

$$\log_3(x+6) + \log_3(x-2) = 3$$

(c) Solve the following equation for x, giving your answer in the form $x = \log_a b$ where $a, b \in \mathbb{N}$:

$$3^{2x-1} = 81$$

Also find the exact value of x.

Question 6 **(30 marks)**

(a) The first three terms of an arithmetic sequence are:

$$3k+2,\ 5k-4,\ 7k-1$$

(i) Find the value of k.

(ii) Using your value of k, find the sum of the first 20 terms of the sequence.

(b) Write down, in descending powers of x, the first three terms in the binomial expansion of:

$$(3x + 2)^8$$

Give each term in its simplest form.

Section B	Contexts and Applications	150 marks

Answer **any three** questions from this section.

Question 7 **(50 marks)**

A company produces open-topped cylindrical containers (no lid). Each container must hold a volume of 400π cm^3.

(a) Let r cm be the radius and h cm be the height of a container.

(i) Show that the total surface area S cm^2 (base and curved side) satisfies:

$$S(r) = \pi r^2 + \frac{800\pi}{r}$$

(ii) Find $S'(r)$.

(iii) Find the value of r that gives the minimum surface area. Verify it is a minimum using $S''(r)$.

(iv) Find the minimum surface area, correct to the nearest cm^2.

(b) A particle moves along a straight line. Its displacement s metres from a fixed point O at time t seconds is:

$$s(t) = 2t^3 - 9t^2 + 12t - 1, \quad t \geq 0$$

(i) Find the velocity $v(t)$ and the acceleration $a(t)$ of the particle.

(ii) Find the times when the particle is instantaneously at rest.

(iii) Find the total distance travelled by the particle in the first 3 seconds.

Question 8 **(50 marks)**

(a) A function f is defined for $x \in \mathbb{R}$, $x \neq 0$ and $x \neq 4$, by:

$$f(x) = \frac{x^2 - 16}{x^2 - 4x}$$

(i) Show that $f(x)$ can be written in a simplified form. Clearly state which value of x must be excluded.

(ii) Find $\lim_{x \to \infty} f(x)$ using the simplified form. Interpret this result geometrically in terms of the behaviour of the graph of $y = f(x)$.

(iii) Using the simplified form, show that $f(x)$ is a decreasing function for all x in its domain.

(b) A ball is dropped from a height of 4 m onto a hard floor. After each bounce, it rises to $\frac{3}{4}$ of the height from which it last fell.

(i) Write down the heights reached after the 1st, 2nd, and 3rd bounces.

(ii) Show that the heights form a geometric sequence and write down the common ratio r.

(iii) Find the height reached after the nth bounce, giving your answer in terms of n.

(iv) Find the total distance travelled by the ball from the moment it is dropped until it theoretically comes to rest.

Question 9 **(50 marks)**

Ciaran and Siobhan take out a mortgage to purchase a home.

(a) They borrow €320,000 at an annual interest rate of 3.6%, compounded monthly.

(i) Write down the monthly interest rate as a decimal.

(ii) They make equal monthly repayments of €1,455.96 at the end of each month, beginning one month after taking out the mortgage.
Write down the present value of each of the first three monthly repayments at the time they take out the mortgage. Give each value as a fraction. Do not multiply out any powers.

(iii) Hence show that the present value A of all n monthly repayments satisfies:

$$A = 1455.96 \times \frac{1 - (1.003)^{-n}}{0.003}$$

(b) Use the formula from part (a)(iii) to find the number of monthly repayments needed to fully repay the mortgage. Give your answer in whole months.

(c) Siobhan's mother Máire invests €800 per month into a savings account at a monthly interest rate of 0.3%. She makes her first deposit now.
Show that the total value of all 12 deposits at the end of 12 months is approximately €9,891.

Question 10 **(50 marks)**

(a) Prove by mathematical induction that, for all $n \in \mathbb{N}$:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

(b) A spherical balloon is being inflated. Air is pumped in at a rate of 50 $cm^3 s^{-1}$.
The volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

(i) Find $\frac{dV}{dr}$.

(ii) Find the rate at which the radius is increasing at the instant when $r = 5$ cm.

(iii) The surface area of a sphere is $A = 4\pi r^2$.
Find the rate at which the surface area is increasing when $r = 5$ cm.

Extra Work Space

Label any work used here clearly with the question number and part.